

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – XI

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Class: TYIT	Batch: 01
Date of Assignment: 01/09/2026	Date/Time of Submission:01/09/2026

Part B – Program

Implementing Advanced Visualization Techniques and Customizations

A) Create Slope Charts and Lollipop Charts

1. Slope Chart

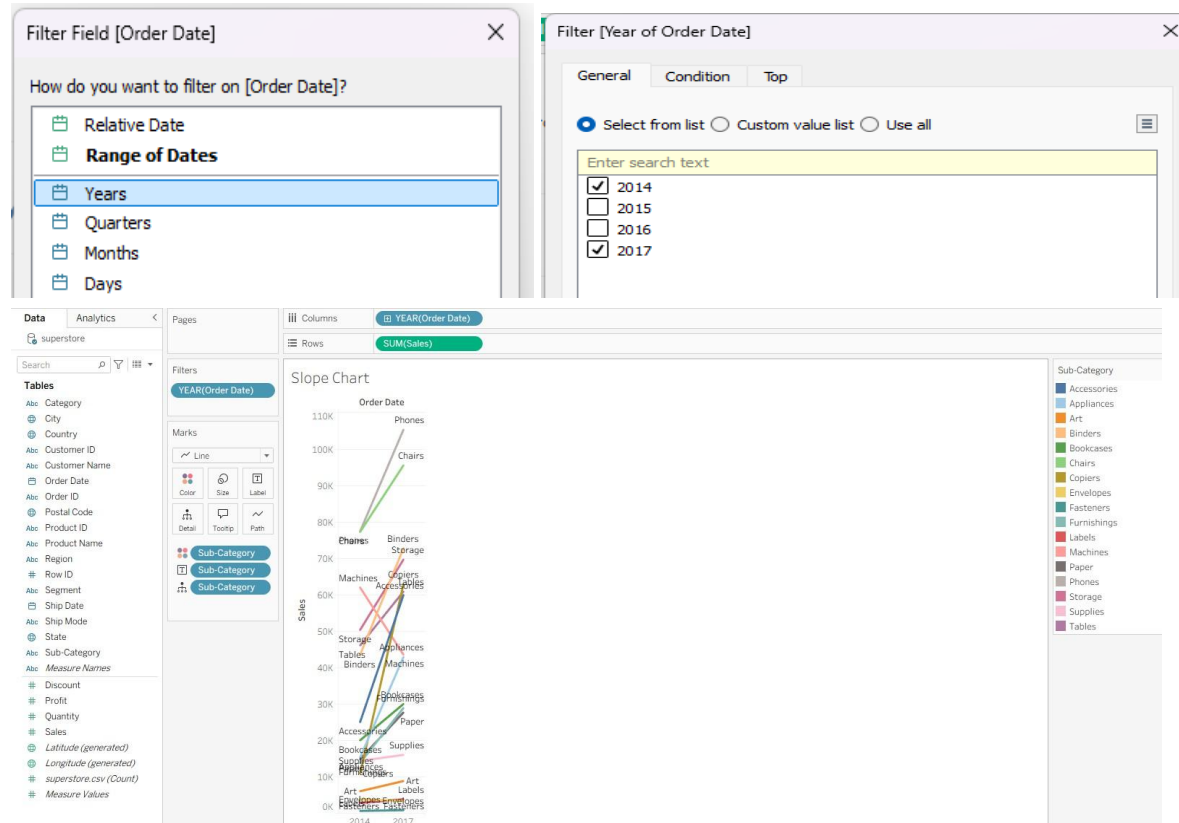
Objective:

To create a slope chart comparing total sales of each Sub-Category between 2014 and 2017.

Steps:

1. Connect Superstore.csv to Tableau and create a worksheet named **Slope Chart**.
2. Drag **Order Date** to Columns and **Sales** to Rows.
3. Add **Order Date** to Filters and select only **2014 and 2017**.
4. Convert YEAR(Order Date) on Columns to **Discrete**.
5. Change the Marks type to **Line**.
6. Drag **Sub-Category** to **Detail** and **Color**.
7. Open **Label**, enable **Show mark labels**, and select **Line Ends**.
8. Drag **Sub-Category** to Label.

Result:



2. Lollipop Chart

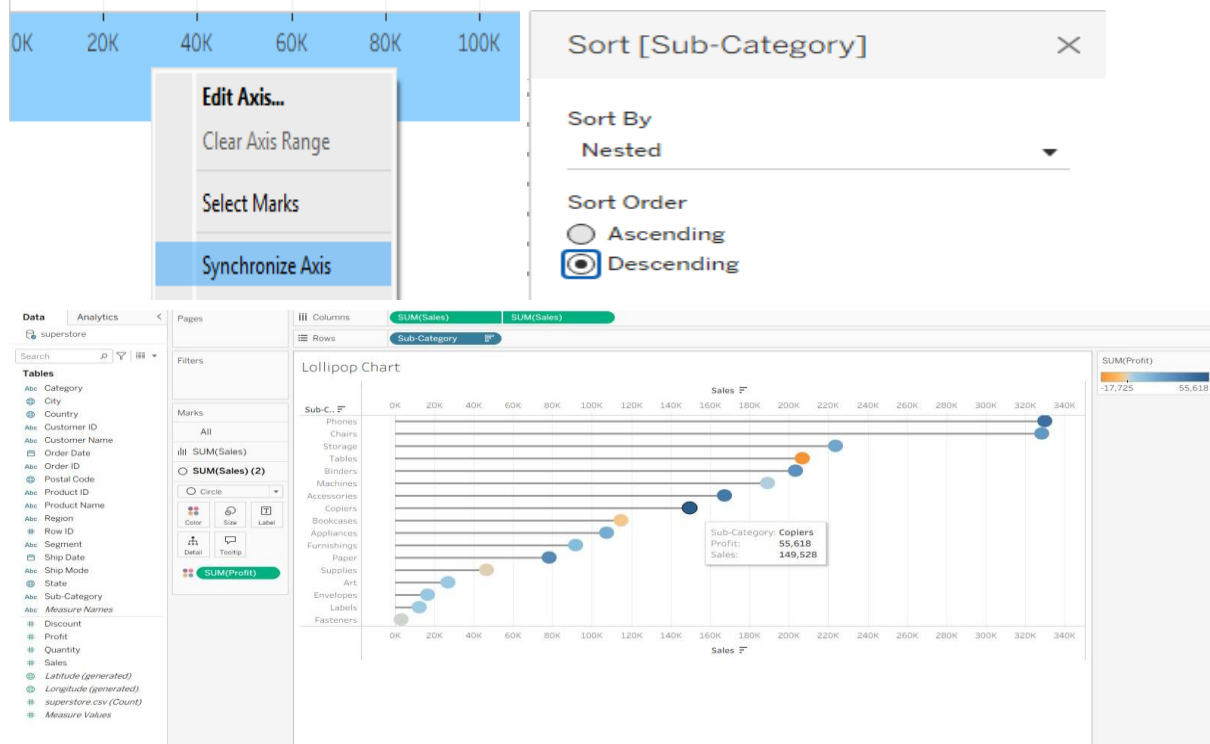
Objective:

To create a lollipop chart showing Sales for each Sub-Category and highlighting Profit using color.

Steps:

1. Create a worksheet named **Lollipop Chart**.
2. Drag **Sub-Category** to Rows.
3. Drag **Sales** to Columns twice.
4. Right-click the second Sales pill and select **Dual Axis**.
5. Select **Synchronize Axis**.
6. Set the first Marks card to **Bar** and reduce its size to create a thin stem.
7. Set the second Marks card to **Circle** and increase its size.
8. Drag **Profit** to **Color** on the Circle Marks card.
9. Sort the Sub-Categories by Sales in descending order.

Result:



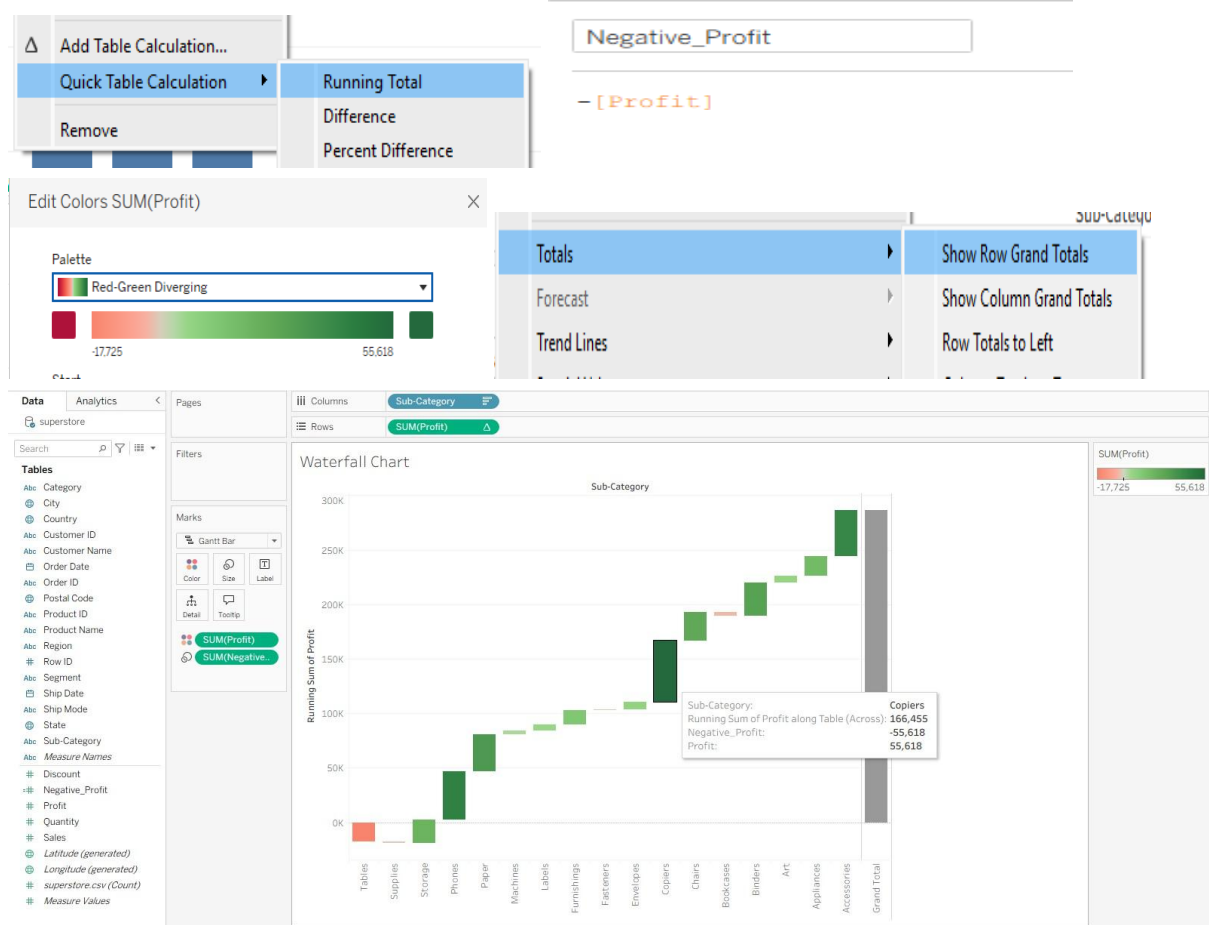
B) Design Waterfall Chart

Objective:

To visualize the cumulative contribution of different Sub-Categories to the overall Profit.

Steps:

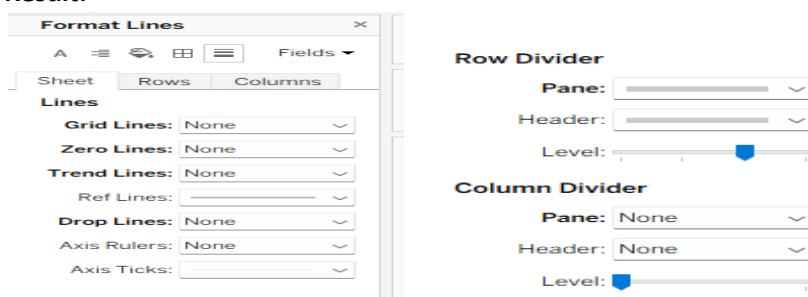
1. Create a worksheet named **Waterfall Chart**.
2. Drag **Sub-Category** to Columns and **Profit** to Rows.
3. Sort the Sub-Categories in descending order.
4. Right-click SUM(Profit) → **Quick Table Calculation** → **Running Total**.
5. Change the Marks type to **Gantt Bar**.
6. Create a calculated field named **Negative_Profit** using:
-[Profit]
7. Drag **Negative_Profit** to **Size**.
8. Drag **Profit** to **Color**.
9. Select **Edit Colors** and choose a **Red-Green Diverging** palette with 0 as the center.
10. Go to **Analysis** → **Totals** → **Show Row Grand Totals**.

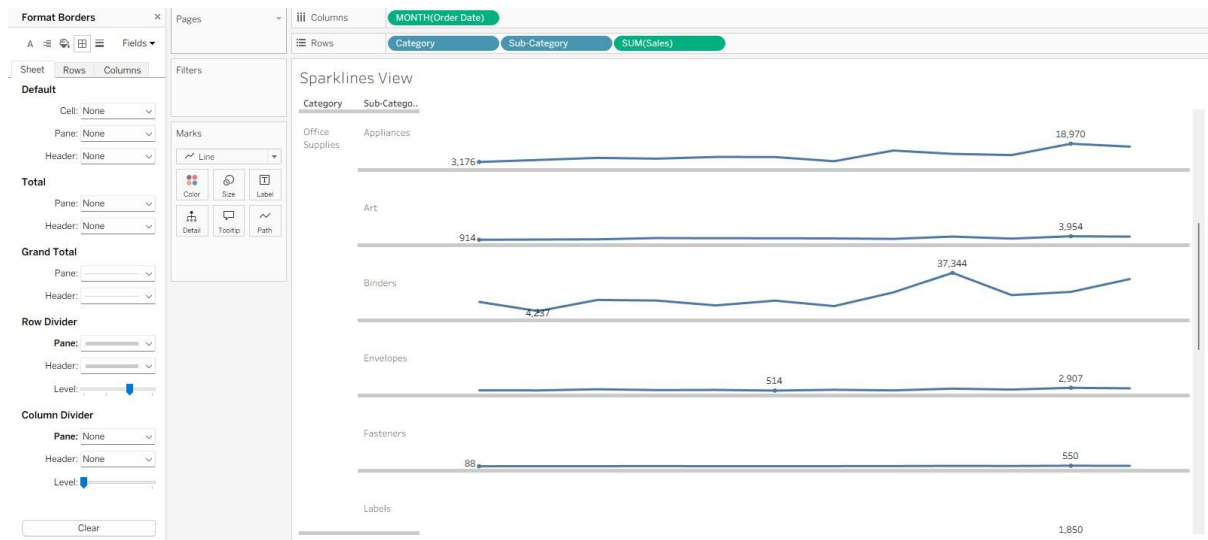
Result:**C) Build Sparklines for Trend Analysis****Objective:**

To create compact sparkline charts showing the monthly Sales trend for each Sub-Category.

Steps:

1. Connect Superstore.csv to Tableau and create a worksheet named **Sparklines View**.
2. Drag **Category** and **Sub-Category** to Rows.
3. Drag **Order Date** to Columns and select **Continuous Month**.
4. Drag **Sales** to Rows and set the Marks type to **Line**.
5. Select **Independent axis ranges for each row or column** from the Sales axis settings.
6. Hide the Sales and Order Date headers.
7. Remove Grid Lines, Zero Lines, Trend Lines, and Drop Lines from the Format pane.
8. Set a thin Row Divider and remove the Column Divider.
9. Enable **Show mark labels** and select **Min/Max** to display the minimum and maximum Sales values.

Result:



D) Implement Sheet Swapping for Dynamic Dashboards

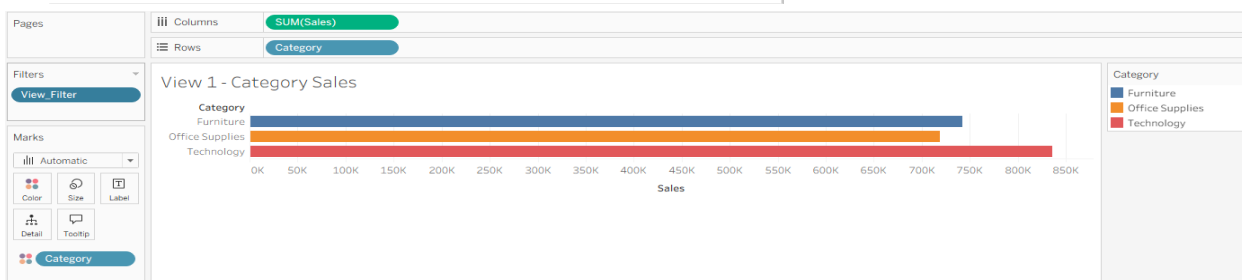
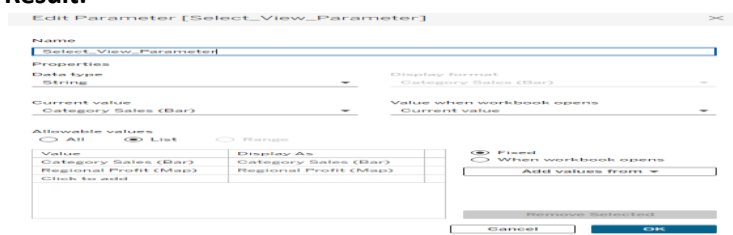
Objective:

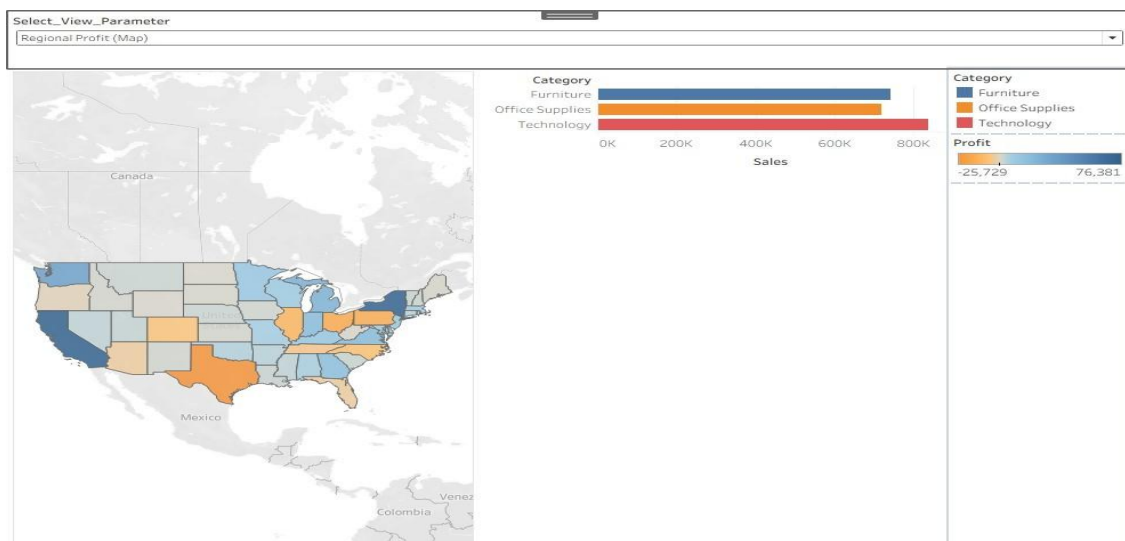
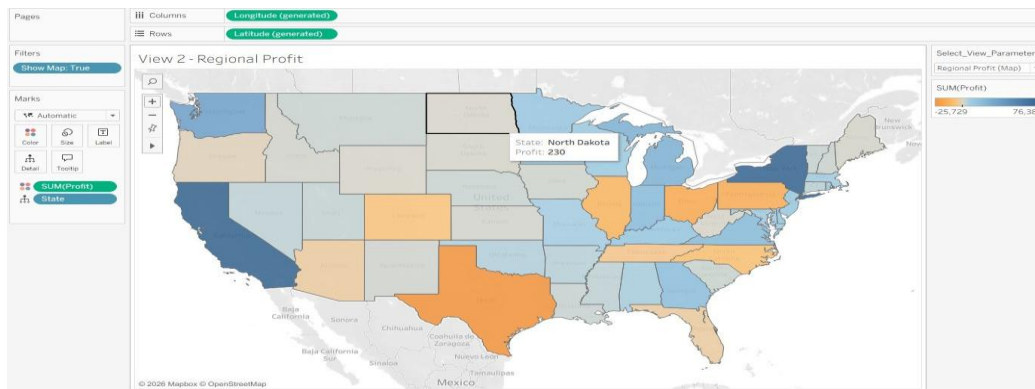
To create a dynamic dashboard that switches between a Category Sales bar chart and a Regional Profit map using a parameter.

Steps:

1. Create a String parameter named **Select_View_Parameter**.
2. Add the values **Category Sales (Bar)** and **Regional Profit (Map)**.
3. Create a calculated field named **View_Filter** using:
[Select_View_Parameter]
4. Create a worksheet named **View 1 - Category Sales**.
5. Drag **Category** to Rows and **Sales** to Columns.
6. Add **View_Filter** to Filters and select **Category Sales (Bar)**.
7. Create another worksheet named **View 2 - Regional Profit**.
8. Create a map using **State** and drag **Profit** to Color.
9. Add 'View_Filter**' to Filters and select **Regional Profit (Map)**.
10. Create a new Dashboard and add a **Vertical Layout Container**.
11. Place both worksheets inside the same container and hide their titles.
12. Add the **Select_View_Parameter** control to the dashboard.
13. Test both parameter options to switch between the two worksheets.

Result:





E) Use Background Images in Dashboards

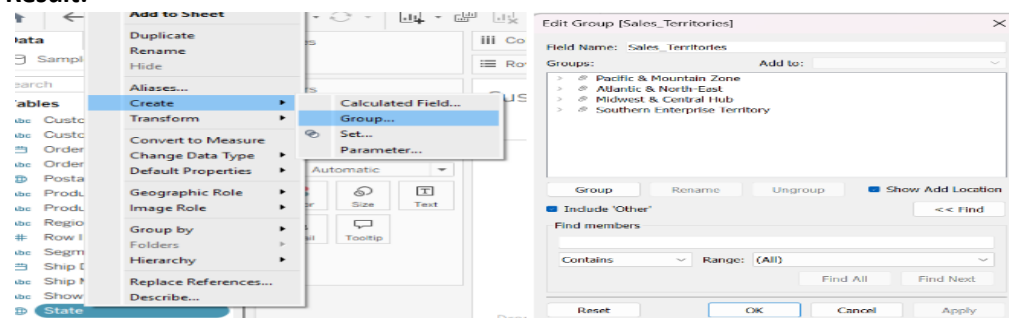
Objective:

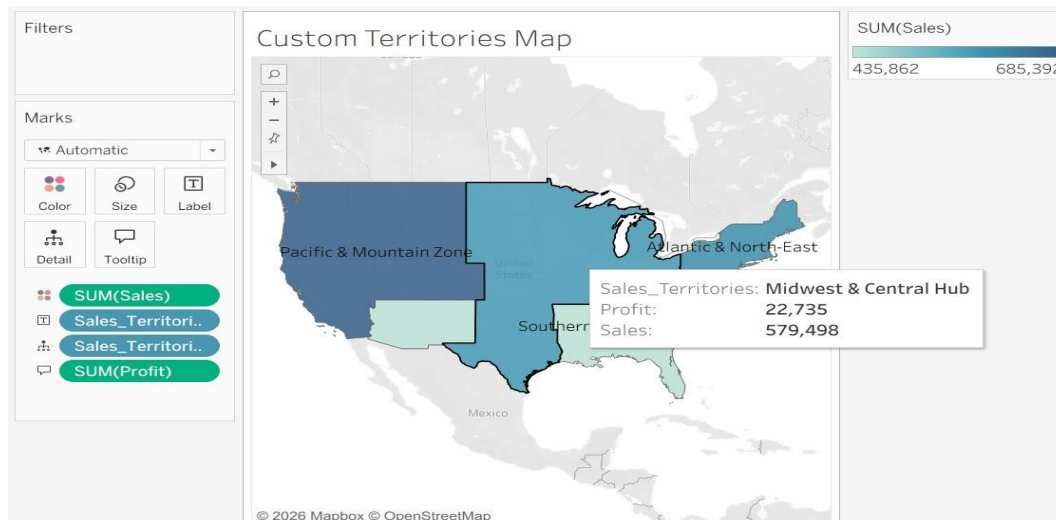
To create a visualization using a background image and coordinate fields.

Steps:

1. Create Store_X_Coordinate and Store_Y_Coordinate calculated fields.
2. Go to **Map** → **Background Images** and add the floor-plan image.
3. Configure the X and Y coordinate fields and image boundaries.
4. Drag Store X to Columns and Store Y to Rows.
5. Add Sub-Category to Detail and Label.
6. Add Sales to Size and Profit to Color.
7. Hide both axes and remove gridlines.

Result:





F) Apply Animation and Transparency Techniques

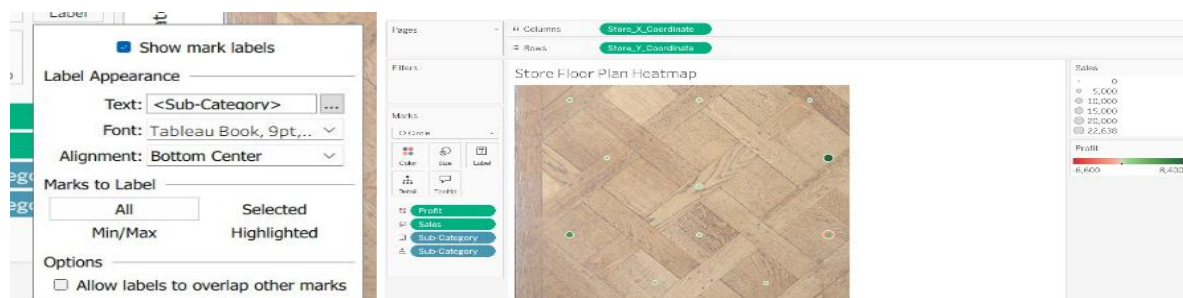
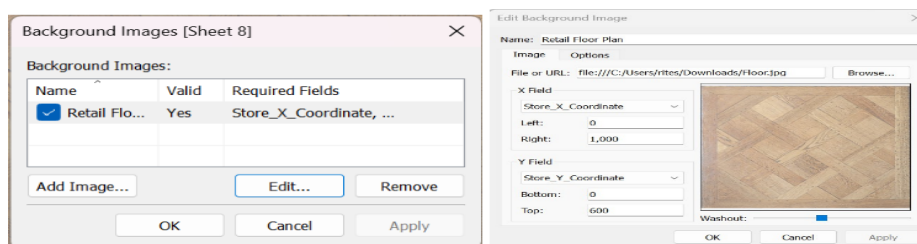
Objective:

To apply animation and transparency techniques for creating an interactive dashboard.

Steps:

1. Create a scatter plot using Sales, Profit and Sub-Category.
2. Add Order Date to the **Pages** shelf and enable **Show History**.
3. Enable **Workbook Animations** and set the duration.
4. Create a KPI card displaying Sales as text.
5. Remove worksheet and pane shading to make the KPI card transparent.
6. Create a dashboard and add the charts as floating objects.
7. Play the animation and verify the motion trails and KPI card.

Result:



G) Apply Animation and Transparency Techniques

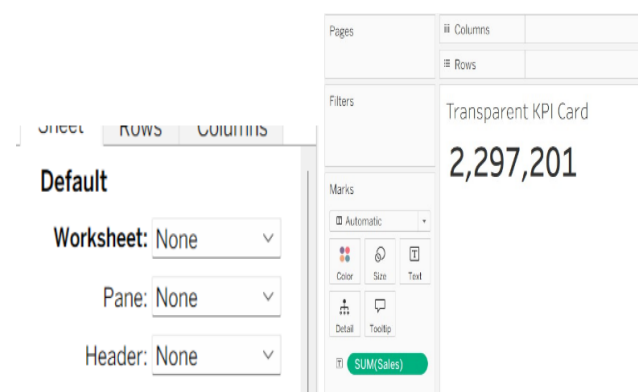
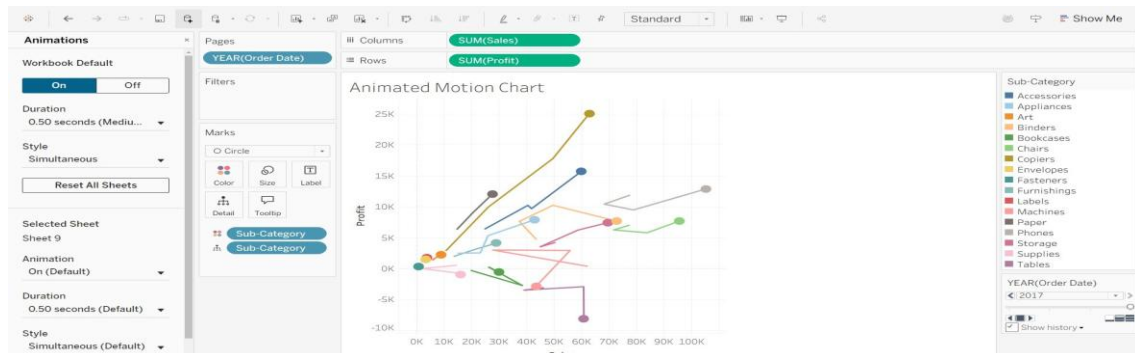
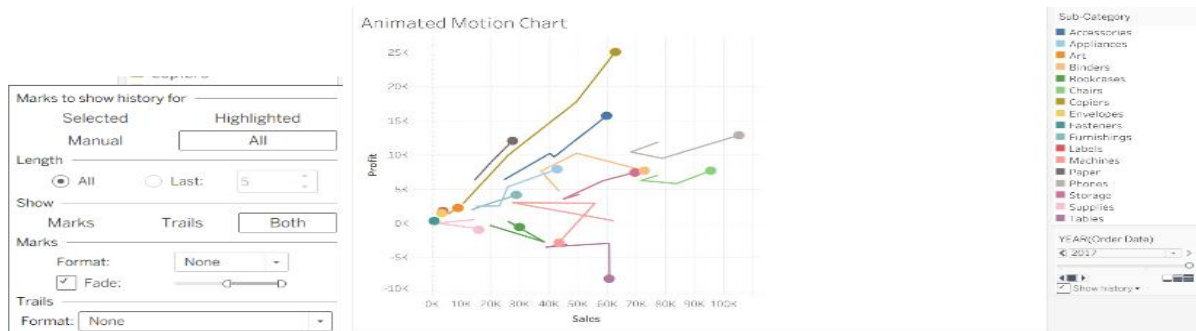
Objective:

To apply animation and transparency techniques to create an interactive Tableau dashboard.

Steps:

1. Create a scatter plot using Sales, Profit and Sub-Category.
2. Add Order Date to the Pages shelf and enable **Show History**.
3. Set the required motion history and playback speed.
4. Enable **Workbook Animations** from Format → Animations.
5. Create a KPI card displaying Sales as text.
6. Remove worksheet and pane shading to make the KPI card transparent.
7. Create a dashboard and add the scatter plot and KPI card as floating objects.
8. Play the animation and verify the motion trails and transparent KPI card.

Result



Objects

Horizontal Container

Vertical Container

A

Text

Extension

Tiled

Floating

☒ Show dashboard title

